

CM0859 – MT5009 Type Theory Constructivism

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2025-2

Preliminaries

Textbook

Pfenning (2023). Lecture Notes on What is (Constructive) Logic?

Source code

The `HASKELL` code have been tested with GHC 9.12.2.

Preliminaries

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\supset$	(if —, then —)	implication
$\perp$	(falsity)	bottom, falsum
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

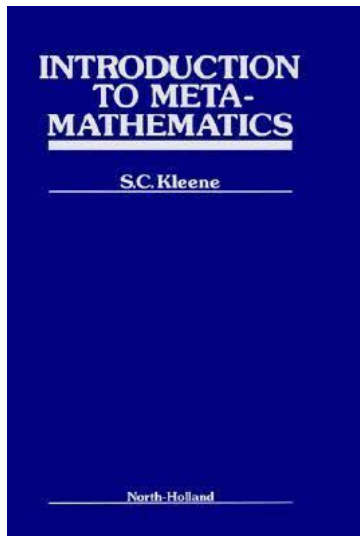
Preliminaries

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\supset$	(if —, then —)	implication
$\perp$	(falsity)	bottom, falsum
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

Definition

We define negation by $\neg A := A \supset \perp$.



The Crisis in the Foundations of Mathematics

Paradoxes $\Rightarrow$ Crisis $\Rightarrow$ $\left\{ \begin{array}{ll} \text{Logicism} & (\text{Russell and Whitehead}) \\ \text{Formalism} & (\text{Hilbert}) \\ \text{Intuitionism} & (\text{Brouwer}) \end{array} \right.$

The Crisis in the Foundations of Mathematics

Logicism (Russell and Whitehead)

“The logicistic thesis is that mathematics is a branch of logic. The mathematical notions are to be defined in terms of the logical notions. The theorems of mathematics are to be proved as theorems of logic.” (Kleene [1952] 1974, p. 43)

The Crisis in the Foundations of Mathematics

Formalism (Hilbert)

“Classical mathematics shall be formulated as a formal axiomatic theory, and this theory shall be proved to be consistent, i.e. free from contradiction.”

(Kleene [1952] 1974, p. 53)

The Crisis in the Foundations of Mathematics

Intuitionism (Brouwer)

“Intuitionism is based on the idea that mathematics is a creation of the mind. The truth of a mathematical statement can only be conceived via a mental construction that proves it to be true.” (Iemhoff 2024)

The Crisis in the Foundations of Mathematics

Conceptions of the infinite

(i) Non-Intuitionism

*“The infinite is treated as **actual** or **completed** or **extended** or **existential**. An infinite set is regarded as existing as a completed totality, prior to or independently of any human process of generation or construction, and as though it could be spread out completely for our inspection.”*
(Kleene [1952] 1974, p. 48)

(ii) Intuitionism

*“The infinite is treated only as **potential** or **becoming** or **constructive**. The recognition of this distinction, in the case of infinite magnitudes, goes back to Gauss, who in 1831 wrote, ‘I protest ... against the use of an infinite magnitude as something completed, which is never permissible in mathematics.’ (Werke VIII p. 216.)”* (Kleene [1952] 1974, p. 48)

Some Differences with Classical Logic

(i) Rejection of the principle of exclude middle (*tertium non datur*).

$\vdash A \vee \neg A$, for all formula A .

Some Differences with Classical Logic

(i) Rejection of the principle of exclude middle (*tertium non datur*).

$$\vdash A \vee \neg A, \text{ for all formula } A.$$

(ii) A proof of an existential formula $\exists x.A(x)$ must include a **witness** t such as $A(t)$ is true.

(continued on next slide)

Some Differences with Classical Logic

(iii) Rejection of proofs by contradiction

Proof by contradiction
(or *reductio ad absurdum*)

$$\frac{\begin{array}{c} [\neg A] \\ \vdots \\ \perp \end{array}}{A}$$

Proof of negation (Bauer 2017)

$$\frac{\begin{array}{c} [A] \\ \vdots \\ \perp \end{array}}{\neg A}$$

Non-Constructive Proofs

Example

To prove that there are irrational numbers $r, s \in \mathbb{R}$ such that r^s is rational.

Non-Constructive Proofs

Example

To prove that there are irrational numbers $r, s \in \mathbb{R}$ such that r^s is rational.

Proof (using the principle of exclude middle)

(whiteboard)

Non-Constructive Proofs

Example

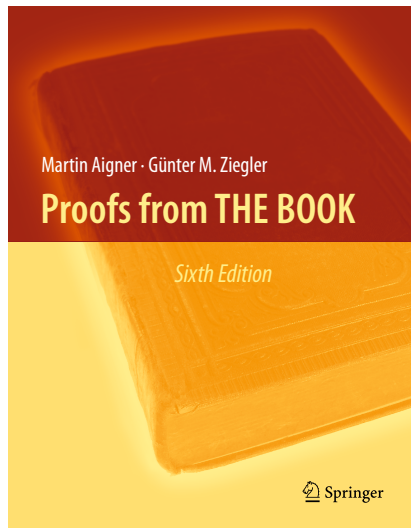
To prove that there are irrational numbers $r, s \in \mathbb{R}$ such that r^s is rational.

Proof (using the principle of exclude middle)

(whiteboard)

Question

Could you give me two irrational numbers r, s such that r^s is rational?



Non-Constructive Proofs

Example

To prove that there are an infinity number of primes.

Non-Constructive Proofs

Example

To prove that there are an infinity number of primes.

Proof (by contradiction)

***“Euclid’s proof.”** For any finite set $\{p_1, \dots, p_r\}$ of primes, consider the number $n = p_1 p_2 \cdots p_r + 1$. This n has a prime divisor p . But p is not one of the p_i : otherwise p would be a divisor of n and of the product $p_1 p_2 \cdots p_r$, and thus also of the difference $n - p_1 p_2 \cdots p_r = 1$, which is impossible. So a finite set $\{p_1, \dots, p_r\}$ cannot be the collection of all prime numbers.”*
(Aigner and Ziegler [1998] 2018, p. 3)

Non-Constructive Proofs

Example

To prove that there are an infinity number of primes.

Proof (by contradiction)

***“Euclid’s proof.”** For any finite set $\{p_1, \dots, p_r\}$ of primes, consider the number $n = p_1 p_2 \cdots p_r + 1$. This n has a prime divisor p . But p is not one of the p_i : otherwise p would be a divisor of n and of the product $p_1 p_2 \cdots p_r$, and thus also of the difference $n - p_1 p_2 \cdots p_r = 1$, which is impossible. So a finite set $\{p_1, \dots, p_r\}$ cannot be the collection of all prime numbers.”*
(Aigner and Ziegler [1998] 2018, p. 3)

Question

Could you give me an infinite list of primes?

Non-Constructive Proofs

Observation

The axiom of choice is a source of non-constructive proofs.

Non-Constructive Proofs

Definition

The **Cartesian product** (or **generalised product**) of a family of sets $\langle A_i \mid i \in I \rangle$ is defined by

$$\prod_{i \in I} A_i := \left\{ f \mid f : I \rightarrow \bigcup_{i \in I} A_i \text{ and } \forall i (i \in I \rightarrow f(i) \in A_i) \right\}.$$

Non-Constructive Proofs

Definition

Axiom of choice: Let $\langle H_i \mid i \in I \rangle$ be a family of sets. If $H(i) \neq \emptyset$ for all $i \in I$, then $\prod_{i \in I} H(i) \neq \emptyset$ (Enderton 1977).

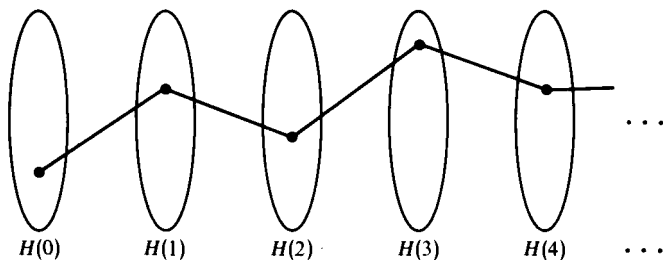


Illustration of the axiom of choice.[†]

[†]Figure source: (Enderton 1977, Fig. 11).

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .
- A proof of $A \vee B$ is either a proof of A or a proof of B together with the information of which of A or B we have a proof.

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .
- A proof of $A \vee B$ is either a proof of A or a proof of B together with the information of which of A or B we have a proof.
- A proof of $A \supset B$ is a function (method, program) which to

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .
- A proof of $A \vee B$ is either a proof of A or a proof of B together with the information of which of A or B we have a proof.
- A proof of $A \supset B$ is a function (method, program) which to
- There is no proof of $\perp$.

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .
- A proof of $A \vee B$ is either a proof of A or a proof of B together with the information of which of A or B we have a proof.
- A proof of $A \supset B$ is a function (method, program) which to
- There is no proof of $\perp$.
- A proof of $\neg A$ is a function (method, program) which transforms a (hypothetical) proof of A into a contradiction.

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .
- A proof of $A \vee B$ is either a proof of A or a proof of B together with the information of which of A or B we have a proof.
- A proof of $A \supset B$ is a function (method, program) which to
- There is no proof of $\perp$.
- A proof of $\neg A$ is a function (method, program) which transforms a (hypothetical) proof of A into a contradiction.
- A proof of $\exists x.A$ is a construction of a witness d and a proof of $A(d)$.

The Brouwer-Heyting-Kolmogorov (BHK) Interpretation

Constructive interpretation of the logical constants

- A proof of $A \wedge B$ is a pair whose first component is a proof of A and whose second component is a proof of B .
- A proof of $A \vee B$ is either a proof of A or a proof of B together with the information of which of A or B we have a proof.
- A proof of $A \supset B$ is a function (method, program) which to
- There is no proof of $\perp$.
- A proof of $\neg A$ is a function (method, program) which transforms a (hypothetical) proof of A into a contradiction.
- A proof of $\exists x.A$ is a construction of a witness d and a proof of $A(d)$.
- A proof of $\forall x.A$ is a function (method, program) which takes an arbitrary individual d into a proof of $A(d)$.

Connection Between Proofs and Programs

Example

We define the follow predicates on natural numbers:

$$\text{even}(x) := \exists y. x = 2y,$$

$$\text{odd}(x) := \exists y. x = 2y + 1.$$

Prove that $\forall x. \text{even}(x) \vee \text{odd}(x)$.

Proof (by induction on x)

(whiteboard)

Connection Between Proofs and Programs

Proof (by induction on x)

(i) **Basis step:** $x = 0$.

Then $\text{even}(x)$ is true because for $y = 0$ (witness), $x = 2y$.

(ii) **Inductive step:** $x = x' + 1$.

For inductive hypothesis $\text{even}(x') \vee \text{odd}(x')$ is true.

- **Case:** $\text{even}(x')$ is true.

That is, $x' = 2y'$ for some y' . Then $x = 2y' + 1$ and therefore $\text{odd}(x)$ is true and the witness is y' .

- **Case:** $\text{odd}(x')$ is true.

That is $x' = 2y' + 1$ for some y' . Then $x = (2y' + 1) + 1 = 2(y' + 1)$ and therefore $\text{even}(x)$ is true and the witness is $y' + 1$.

Connection Between Proofs and Programs

HASKELL function “from” the proof that $\forall x.\text{even}(x) \vee \text{odd}(x)$

```
1  data Nat = Zero | Succ Nat
2
3  data E0 = Even | Odd
4    deriving Show
5
6  isEvenOrOdd :: Nat -> E0
7  isEvenOrOdd Zero      = Even
8  isEvenOrOdd (Succ n) = case isEvenOrOdd n of
9    Even -> Odd
10   Odd  -> Even
```

Connection Between Proofs and Programs

HASKELL function with witness “from” the proof that $\forall x.\text{even}(x) \vee \text{odd}(x)$

```
1  data Nat = Zero | Succ Nat
2      deriving Show
3
4  data E0 = Even Nat | Odd Nat
5      deriving Show
6
7  isEvenOrOdd :: Nat -> E0
8  isEvenOrOdd Zero      = Even Zero
9  isEvenOrOdd (Succ x) = case isEvenOrOdd x of
10      Even y -> Odd y
11      Odd y  -> Even $ Succ y
```

References

-  Martin Aigner and Günter M. Ziegler [1998] (2018). Proofs from THE BOOK. 6th ed. Springer. DOI: doi.org/10.1007/978-3-662-57265-8 (cit. on pp. 18–20).
-  Andrej Bauer (2017). Five States of Accepting Constructive Mathematics. Bulletin of the American Mathematical Society 54.3, pp. 481–498. DOI: [10.1090/bull/1556](https://doi.org/10.1090/bull/1556) (cit. on p. 13).
-  Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 23).
-  Rosalie Iemhoff (2024). Intuitionism in the Philosophy of Mathematics. In: The Stanford Encyclopedia of Philosophy. Ed. by Edward N. Zalta and Uri Nodelman. Summer 2024. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/sum2024/entries/intuitionism> (cit. on p. 9).
-  Stephen Cole Kleene [1952] (1974). Introduction to Metamathematics. Seventh reprint. Bibliotheca Mathematica: A Series of Monographs on Pure and Applied Mathematics. Volume 1. North-Holland (cit. on pp. 7, 8, 10).
-  Frank Pfenning (17th Jan. 2023). Lecture Notes on What is (Constructive) Logic? Material for the course 15-317 Constructive Logic at Carnegie Mellon University, Spring 2023. URL: <https://www.cs.cmu.edu/~fp/courses/15317-s23/> (cit. on p. 2).